

Joint Detection of Change Points in Multichannel Single-Molecule Measurements

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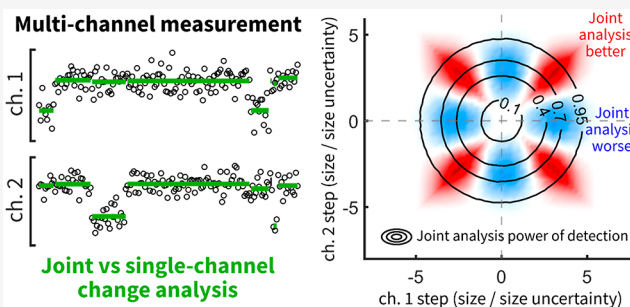
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ABSTRACT: Recent developments in single-molecule measurement technology have expanded the capability to measure multiple parameters. These emergent modalities provide more holistic observations of complex biomolecular processes and call for new analysis methods to detect state changes in multichannel data. Here we develop an algorithm called MULLR (Multi-channel Log-Likelihood Ratio test) to *jointly* identify change points in multichannel single-molecule measurements. MULLR is an extension of the popular single-channel implementation for change point detection based on a binary segmentation and log-likelihood ratio test framework. We validate the algorithm on simulated data and characterize the power of detection and false positive rate. We show that MULLR can identify change points in experimental multichannel data and naturally works with different noise statistics and time resolutions across channels. Further, we quantify the benefit of MULLR compared to single-channel analysis. We envision that the MULLR algorithm will be useful to a range of multiparameter single-molecule measurements.



INTRODUCTION

Single-molecule investigations of biomolecules can provide rich access to unsynchronized dynamics obscured in ensemble measurements.^{1,2} A hallmark of these single-molecule measurements is abrupt changes in the measured time trace, indicative of state transitions of the molecule of interest. These transitions are of particular interest because they yield mechanistic information not available in traditional ensemble experiments. However, due to the often limited signal-to-noise ratio in single-molecule recordings, robust detection of state transitions ("change points") can be challenging. Early single-molecule measurements relied on visual inspection³ or simple techniques such as thresholding.⁴ Since then, more sophisticated tools, which model the statistical properties of single-molecule measurements, have been developed. Approaches include (but are not limited to) hidden Markov models (HMMs),^{5–11,34} Bayesian inference,^{7,8,11} application of a likelihood ratio test,^{12–14} or information-based approach^{15,16} together with a binary segmentation procedure and, recently, deep learning.¹⁷ Importantly, most of these approaches focus on time traces of a single measurement quantity only (for example, smFRET efficiency).

Recently, to better understand biological processes of increasing complexity, many "hybrid" single-molecule techniques have emerged.¹⁸ They combine different modalities to simultaneously measure two or more parameters of a single molecule of interest (for example, protein conformation

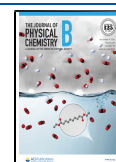
together with substrate DNA unwinding¹⁹). These hybrid approaches provide critical information about correlations between different degrees of freedom of the system, yielding new insight into biological function at the molecular level. We have recently developed ABEL-FRET as another example of a multimodal single-molecule technique.²⁰ ABEL-FRET simultaneously measures the smFRET, diffusivity (D), and electrokinetic mobility (μ) of a single molecule in solution with feedback trapping^{21,22} and can probe the intramolecular conformation together with biomolecular interactions.²⁰ These hybrid single-molecule approaches generate synchronized time traces of multiple measurement quantities and call for new algorithms that *jointly* analyze state changes in a multidimensional parameter space. Several popular analysis approaches can be extended to multiple dimensions but come with limitations (for example, with HMM, a specific kinetic scheme must be assumed).

Here we develop a more general algorithm that jointly detects change points in multichannel single-molecule measurements.

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There are several challenges to overcome: In general, the number of states of the measured system, the kinetic scheme, and the location of each state in the parameter space are not known *a priori*. In practice, parallel measurement channels can have different time resolution and measurement statistics (e.g., Poisson noise for photon counting measurements and Gaussian noise for diffusivity measurements). To address these challenges, we build our algorithm on a likelihood ratio test and binary segmentation framework,^{12–14,23,24} which naturally incorporates measurements with flexible time resolution and noise statistics. Furthermore, it does not require any assumption of a particular kinetic scheme. The one-channel implementation of this framework has been successfully used in a diverse array of single-molecule experiments^{25–27} and recently expanded to estimate both velocity and diffusion coefficient changes in single-particle tracking trajectories.²⁴ In this manuscript, we first describe our Multi-channel Log-Likelihood Ratio test (MULLR) algorithm as an extension of the original single-molecule change point analysis work¹² to jointly identify change points across parallel measurements. We then validate the MULLR algorithm on simulated data and characterize the power of detection under a controlled false positive rate. We next show that our algorithm works on experimental data from the ABEL-FRET platform. Furthermore, with multiple measurement channels, there is a choice of whether to analyze data jointly using the algorithm developed in this work or to analyze the channels individually.²⁸ We show that, for a simulated case of two Gaussian channels, the benefit of joint detection using MULLR compared to a per-channel analysis depends on the degree of correlation between the two channels as well as the signal-to-noise ratio on each channel. These results serve as a starting point to discuss optimal data analysis strategies for hybrid single-molecule techniques with multiple synchronized readouts.

METHODS

Generation of Simulated Traces. All simulated traces were generated using custom scripts written in MATLAB. Simulations were run on either a Windows 10 desktop computer (Windows Enterprise Version 10.0 (Build 19042)) running MATLAB Version 9.7.0.1216025 (R2019b) or the cetus computing cluster of the Lewis-Sigler Institute (Springdale Linux 7.9 (Verona)) running MATLAB Version 9.4.0.813654 (R2018a). In brief, a simulation is set up by choosing for each Poisson channel a mean for each time bin and for each Gaussian channel a mean and standard deviation for each time bin. The mean values are chosen based on the desired dynamics of the underlying system during the simulation (e.g., a constant level, two constant levels separated by a single change point, multiple transitions back and forth between two levels). A specific simulated trace is then generated by drawing from a Gaussian (or Poisson) distribution with the appropriate parameters for each time bin. More details for each set of simulations discussed in the results can be found in the [Supporting Methods](#) section of the [Supporting Information](#).

Determining the LLR Threshold Corresponding to a Certain False Positive (Type I Error) Rate. In MULLR, the peak of a log-likelihood ratio (LLR) profile is only accepted as a change point if the LLR value at the peak exceeds a certain threshold. Intuitively, applying an LLR threshold balances the false positive rate and the power of detection of true change points. If the threshold is small, peaks arising from random fluctuations in the LLR profile may give rise to false positives. On

the other hand, a high threshold makes sure almost every identified change point is true but may miss real change points (i.e., a low power of detection). MULLR follows a rigorous statistical procedure developed in refs 12 and 29 to compute the LLR threshold with a user-provided false positive rate (α). Specifically, define $Z_N = \max_{1 \leq k < N} (2\mathcal{L}_{k,\text{tot}})$, which corresponds to twice the maximum of the LLR profile. Then, the critical threshold value for $Z_N^{1/2}$ is defined by

$$\tilde{z}_N = \max(x: P\{Z_N^{1/2} \leq x\} \leq 1 - \alpha)$$

Vostrikova²⁹ showed that we can calculate the critical value $\tilde{z}_N$ by finding the roots of the following equation (in x)

$$\frac{x^d \exp(-x^2/2)}{2^{d/2} \Gamma(d/2)} \left\{ T - \frac{d}{x^2} T + \frac{4}{x^2} + o\left(\frac{1}{x^4}\right) \right\} = \alpha$$

where $\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx$ is the gamma function, d is the number of parameters which can change at the change point (e.g., $d = 2$, if we have two Gaussian channels where the mean can change but the variance is constant), α is the allowed rate of false positives given by the user, N is the number of data points, and $T = \log\left(\frac{(1-h)^2}{h^2}\right)$, with $h(N) = \frac{(\log N)^{3/2}}{N}$, where we choose the form of $h(N)$ following Gombay and Horváth.³⁰

This threshold ($\tilde{z}_N$) for $Z_N^{1/2}$ can be converted into the threshold used in the MULLR algorithm (r_N) for the maximum log-likelihood ratio $\max_{1 \leq k < N} (\mathcal{L}_{k,\text{tot}})$ using the relation

$$r_N = \frac{\tilde{z}_N^2}{2}$$

If $\max_{1 \leq k < N} (\mathcal{L}_{k,\text{tot}}) > r_N$, we accept the change point, and if $\max_{1 \leq k < N} (\mathcal{L}_{k,\text{tot}}) < r_N$, we reject the change point. Importantly, the value of this significance threshold depends on N , the number of bins in the time trace segment, α , the user supplied allowed rate of false positives, and d , the number of parameters which can change at the change point. The threshold must be recalculated during every LLR test step due to the dependence on N . More details can be found in the [Supporting Results](#) section of the [Supporting Information](#).

Transition Step Size and Uncertainty. In the manuscript, we consider dynamic and noisy time traces consisting of constant levels separated by abrupt changes. We need to define a measure of the magnitude and uncertainty in each step size. For a time trace x_t with N bins, constant measurement noise (standard deviation) σ_n , and a step after the k th bin, the step size can be quantified as

$$S_{\text{step}} = \bar{x}_{k+1,N} - \bar{x}_{1,k}$$

where $\bar{x}_{i,j}$ represents the mean of measured x_t values in the interval $[i, j]$. To quantify the uncertainty in S_{step} , we first consider the uncertainty in $\bar{x}_{i,j}$, which is given by the standard error in the mean: $\sigma_{\bar{x}_{i,j}} = \frac{\sigma_n}{\sqrt{j-i+1}}$. Then, the overall uncertainty in the step size is given by the sum of the variance of each level mean:

$$\sigma_{\text{step}}^2 = \sigma_n^2 \left(\frac{1}{k} + \frac{1}{N-k} \right)$$

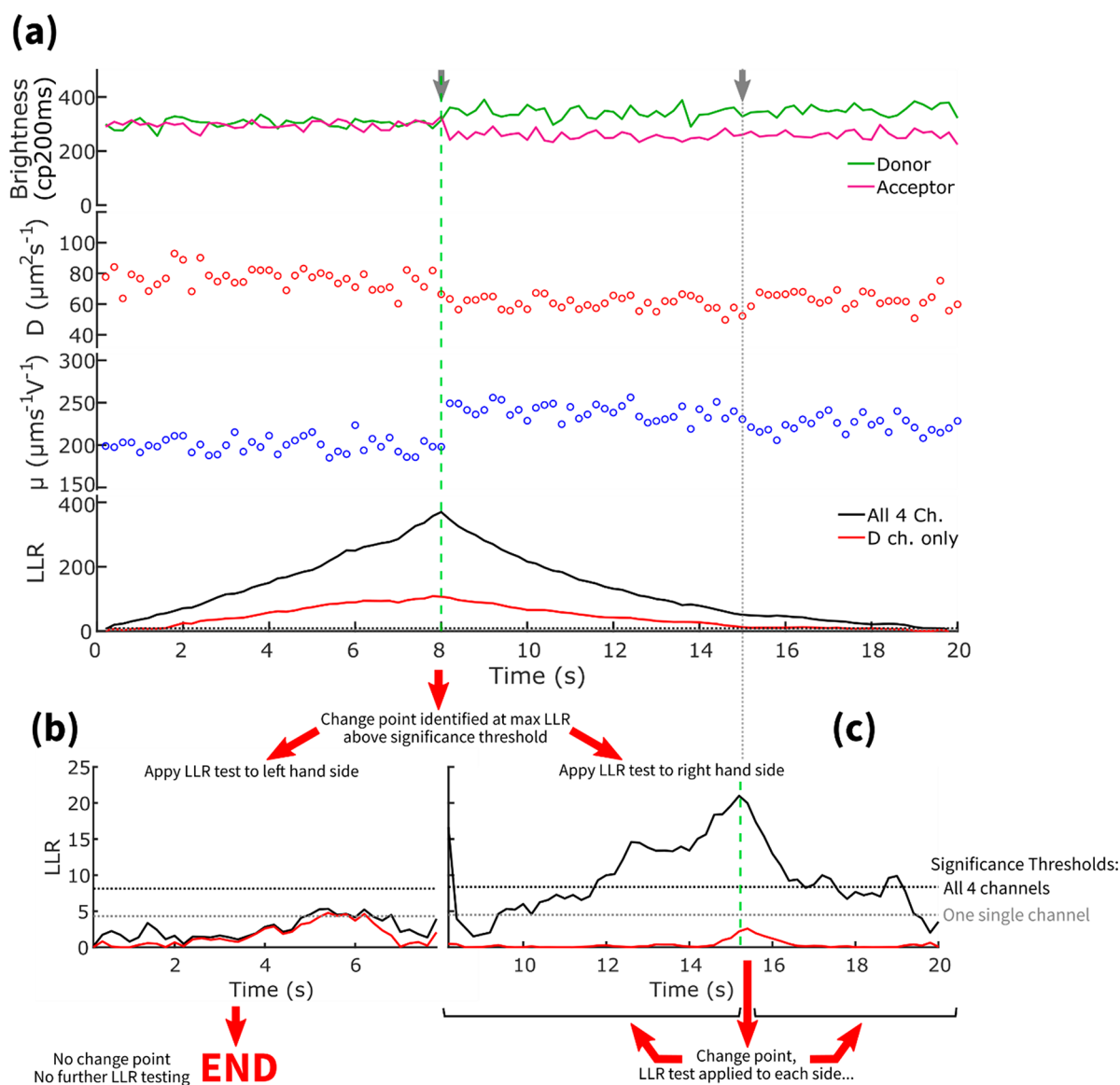


Figure 1. Joint change point detection using the Multi-channel Log-Likelihood Ratio (MULLR) algorithm. (a) Simulated ABEL-FRET data showing two change points (gray arrows) and how they are detected using the MULLR algorithm. The top three panels show simulated time traces of the donor and acceptor brightness, the diffusion coefficient (D), and the electrokinetic mobility (μ). In all panels, the time resolution is 200 ms/bin. The fourth panel shows the LLR profile at the first iteration of the algorithm (black, LLR calculated jointly from all four channels; red, LLR calculated from a single channel of D), which peaks above the significance threshold (dotted horizontal line close to the time axis). The vertical green dashed line denotes the identified change point. The MULLR algorithm subsequently operates on both the “before” (b) and “after” (c) portions of the data in panel a. In panels b and c, significance thresholds are shown for both joint (black dotted line) and single-channel (gray dotted line) test statistics.

The ratio $\frac{S_{\text{step}}}{\sigma_{\text{step}}}$ is an important parameter in determining how easy the step is to detect.

Application of MULLR to Experimental ABEL-FRET Data. The MULLR algorithm must be supplemented with several preprocessing steps to analyze experimental ABEL trap data. First, we have to make sure the fluorescence intensity data (recorded photon by photon), the diffusion coefficient (D), and electrokinetic mobility (μ) data (calculated typically at 100 or 200 ms time resolution) have the same time resolution. In this work, our approach is to bin the fluorescence intensity data with the same time resolution as the D and μ estimation.

Second, an appropriate estimate of the standard deviation in the measured D and μ must be supplied as input to the algorithm. We have previously shown that the measurement

noise in D (and μ) is approximately Gaussian and the measured spread (standard deviation) in D approaches that generated by a realistic ABEL trap simulator.³¹ In this work, we thus use simulated standard deviation values (Supplementary Methods).

RESULTS

Joint Change Point Detection Using the Multi-channel Log-Likelihood Ratio (MULLR) Test. The MULLR algorithm builds on the binary segmentation and log-likelihood ratio test framework, which was first developed for single-molecule data by Watkins and Yang.¹² This previous work operates on a single measurement channel with Poisson statistics. We recap how the algorithm is applied to an N -bin time trace below. Briefly, to identify change points, each bin is evaluated for the possibility of a change point after that bin (for

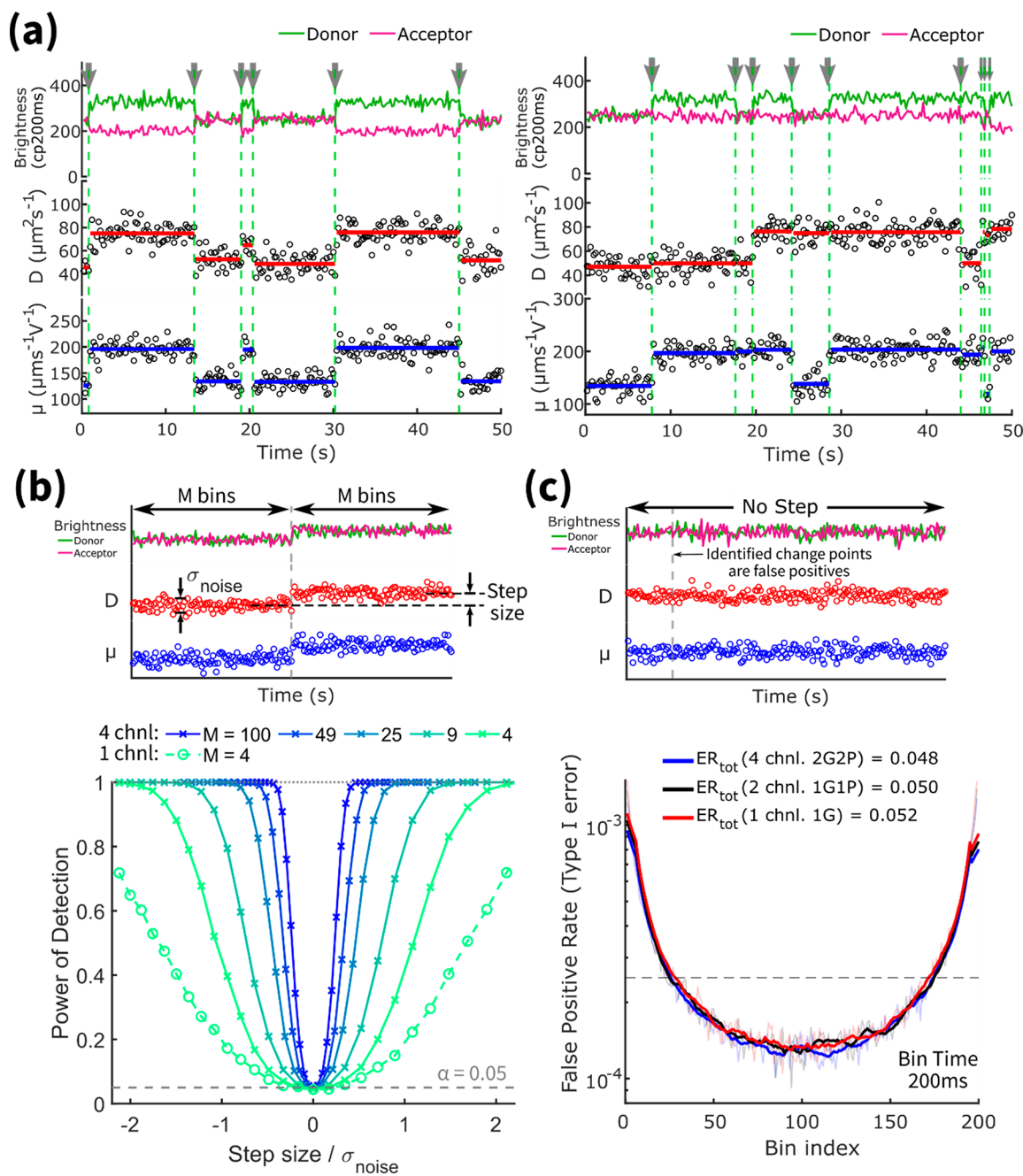


Figure 2. Characterizing MULLR using simulated traces. (a) Example time traces (donor and acceptor channel brightness, diffusion coefficient D , and electrokinetic mobility μ) where each channel can transition between two states. The gray arrows indicate the true change point locations, and the green dashed lines are the change points identified by MULLR. The thick red and blue lines indicate the mean D and μ levels between consecutive change points. (b) Characterizing the power of detection of MULLR. Top: schematic of the simulation setup with the definition of M , step size, and σ_{noise} (the standard deviation of the measurement noise). Bottom: the power of detection as a function of noise-normalized step size. The step size and noise are the same on each channel. The power of detection for a single Gaussian channel with the same step sizes and noise is shown for comparison. The horizontal gray lines are drawn at 0.05 (equal to the false positive rate α , dashed) and 1 (dotted). False positive rate: $\alpha = 0.05$. The symbols give the results from simulations, and the curves provide a guide to the eye. (c) False positive rate (type I error rate) of MULLR. Top: schematic showing the simulation setup. Bottom: False positive rate for each bin for simulated data with 1 (red), 2 (black), and 4 (blue) channels. For each condition (color), the faint curve is the false positive rate for each bin and the solid curve is the 10-point moving average. The horizontal black dashed line indicates the expected false positive rate per bin for false positives uniformly distributed over the trace (i.e., $\alpha/200 = 2.5 \times 10^{-4}$).

bin index $k = 1, 2, \dots, N - 1$). For the k th bin, two hypotheses are considered: the alternative hypothesis (H1: there is a change point between bin k and bin $k + 1$) and the null hypothesis (H0: there is no change point between bin k and bin $k + 1$). The likelihood of the observed data is evaluated under each hypothesis (L_k^A and L_k^0), and the log-likelihood ratio value

($\mathcal{L}_k = \log \frac{L_k^A}{L_k^0}$, LLR) is computed and assigned to the k th bin.

This LLR quantifies the degree to which the alternative hypothesis is favored over the null hypothesis. After the LLR value is computed for all bins ($k = 1, \dots, N - 1$), the peak in the resulting profile of LLR versus the bin index defines the location

of a candidate change point. This change point is accepted if the peak LLR exceeds a significance threshold (r_N , included to control the false positive rate; see [Methods](#)). After a change point is detected (say at bin index $k = j$), the time trace is partitioned into two segments, one before the identified change point and one after. The LLR test is then applied to each of these two new segments. This process of segmentation and LLR testing continues recursively until no further change points are found.

The MULLR algorithm extends the original work by searching for change points across *multiple* measurement channels which can exhibit different statistics. To achieve this, we first bin the data so that all measurement channels have the same time resolution and the same total number of bins. We then replace the single-channel measurement likelihood used in previous works with the joint likelihood of the data over all measurement channels $L_{k,\text{tot}}$. Importantly, we assume that the measurement channels have independent noise, so that the total likelihood function is the product of the likelihood of each individual channel. Specifically, for ABEL-FRET data, which consists of two Poisson channels (donor and acceptor intensities) and two Gaussian channels (diffusivity and electrokinetic mobility), we have $L_{k,\text{tot}} = L_{k,\text{P1}}L_{k,\text{P2}}L_{k,\text{G1}}L_{k,\text{G2}}$ (where the subscript P denotes a Poisson channel and G denotes a Gaussian channel). Note that the log-likelihood ratio (of the alternative over the null hypothesis) can then simply be expressed as a sum of the LLR of the individual measurement channels: $\mathcal{L}_{k,\text{tot}} = \mathcal{L}_{k,\text{P1}} + \mathcal{L}_{k,\text{P2}} + \mathcal{L}_{k,\text{G1}} + \mathcal{L}_{k,\text{G2}}$, where $\mathcal{L}_{k,\text{Pi}}$ (or $\mathcal{L}_{k,\text{Gi}}$) is the log-likelihood ratio of the i th Poisson (or Gaussian) channel. This “total” log-likelihood ratio profile ($\mathcal{L}_{k,\text{tot}}$) remains a one-dimensional function of the binning (or time) index but now incorporates the joint statistics of all measurement channels. As in the single-channel case, MULLR then finds the bin location with the maximum LLR and identifies it to be a change point if the maximum LLR exceeds the LLR threshold determined by a user-supplied false positive rate (α). Note that the proper LLR threshold calculation using the correct number of free parameters across all measurement dimensions needs to be used ([Methods](#) and [Supplementary Methods](#)). Once a change point is identified, the above multichannel log-likelihood ratio test is applied to the trace segments before and after the change point and this process continues recursively on binary segmented portions of the trace until no more change points are found, as in the single-channel case. Finally, in this work, for each Gaussian channel, we supply a predetermined variance to the algorithm (see [Supplementary Methods](#)) instead of extracting it from the data. This is valid when the observed variance originates from measurement noise and does not give useful information about the measured molecules. This also allows us to incorporate our prior knowledge of the expected magnitude of the measurement noise (for example, from realistic simulations). Further details of the MULLR algorithm can be found in [Supplementary Results](#).

[Figure 1](#) illustrates the workflow of MULLR using simulated ABEL-FRET data. Here, the ground truth contains two change points, one at 8 s, where all four measurement channels undergo significant and abrupt changes, and another one at 15 s, where only the D and μ channels undergo more subtle but still digital changes. The first pass of MULLR (panel a) identifies the first major change point (at 8 s), because the joint LLR ratio (black curve in the LLR plot) has a peak that exceeds the threshold computed with a false positive rate of $\alpha = 0.05$ (dashed

horizontal line). Note that, in this case, running the single-channel change point algorithm on only the D channel identifies the same change point (red curve in the LLR plot). Once the first change point is identified, MULLR continues LLR testing on the segmented portions of the trace (panels b and c). In this case, no further change points are found in the “before” segment, because the maximum value of LLR is below the threshold (panel b). On the other hand, the “after” segment yields another change point at 15 s, successfully recovering the ground truth. It is important to note that the single-channel algorithm on D would miss the subtle change point at 15 s ([Figure 1c](#), the peak of the red LLR profile is below the single-channel threshold), suggesting MULLR might take advantage of the multiple measurement channels to enhance the power of detection. For other individual channel LLR profiles, see [Figure S1](#).

Characterization of MULLR. MULLR is based on a “(kinetic) model free” framework. This means it should capture dynamic changes across multiple channels regardless of the underlying dynamics of the system (i.e., the number of states, whether transitions are Markovian or not, etc.). Indeed, we use simulated ABEL-FRET traces to validate that the MULLR algorithm can capture the true change point locations both for synchronized transitions across all channels ([Figure 2a](#), left) and asynchronous changes ([Figure 2a](#), right). At the same time, the algorithm works on states that follow exponentially distributed ([Figure 2a](#)) or non-exponentially distributed ([Figure S2](#)) dwell times.

Next, we characterize the power of detection (or sensitivity), which is the fraction of true change points detected by the algorithm. To this end, we simulated a batch of traces each containing four measurement channels (two Poisson and two Gaussian). For each trace, there is a synchronous step in the middle ([Figure 2b](#), top) across all four channels. Intuitively, the larger the step size and/or the smaller the noise (bin-to-bin standard deviation σ_{noise}), the easier it will be to detect the change point. We thus vary the relative step size (step size divided by σ_{noise} , same for all four channels) and calculate the percentage of successes using MULLR (applied to 10,000 simulated traces) as the power of detection ([Figure 2b](#), bottom). We found that the power of detection shows a minimum at zero step size with a value equal to the allowed rate of false positives ($\alpha = 0.05$).¹² The detection power increases with step size ([Figure 2b](#)), and the rate of increase depends on the number of bins (M) that define a stable level. The larger the value of M , the steeper the increase. This can be understood by considering the uncertainty in the mean of the level before (or after) the change point, which scales as $\sigma_{\text{noise}}/\sqrt{M}$. The power of detection ultimately depends on the size of the step compared to the uncertainty in the size of the step (i.e., the step size to uncertainty ratio, $\frac{s_{\text{step}}}{\sigma_{\text{step}}}$, [Methods](#)). If we consider the step size

relative to its uncertainty, the power of detection curves for different M (almost fully) collapse onto a single curve ([Figure S3](#)). We also found that the power of detection for four channels is much larger than one channel (comparing the solid green curve to the dashed green curve in [Figure 2b](#), bottom, equal step sizes and noise on each channel).

We next characterize the false positive (type I error) rate of the MULLR algorithm using simulations in which there are no true change points. We have configured MULLR to restrict the false positive rate to $\alpha = 0.05$ ([Methods](#)). In this case, all identified change points are false positives and we record their frequency and distribution along the trace. [Figure 2c](#) plots the

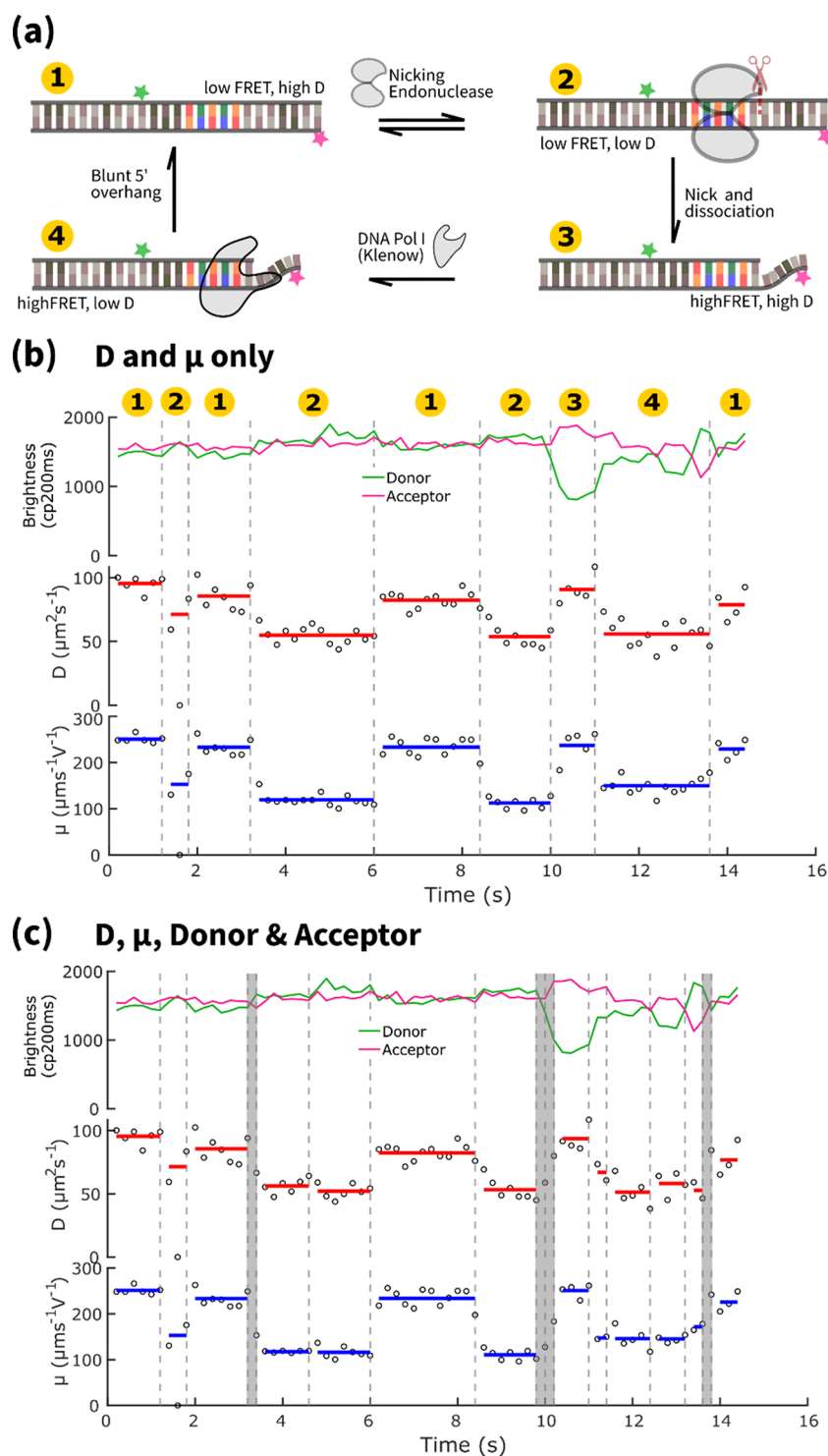


Figure 3. MULLR identifies change points in experimental ABEL-FRET visualization of dynamic DNA damage and repair processes. (a) Experimental scheme: a single DNA molecule goes through continuous nicking (damage) and polymerization (repair) cycles which are monitored using FRET and measurements of the single-complex diffusion coefficient (D) and electrokinetic mobility (μ). (b) Example single-molecule trace in which change points (vertical dashed lines) are identified jointly using D and μ . Numbers above the trace are the state assignments based on the scheme in panel a. The horizontal red and blue lines indicate the mean D and μ values for each level between consecutive change points. False positive rate: $\alpha = 0.05$. (c) The same molecule as in panel b but with change points identified jointly using all four channels (donor, acceptor brightness, D , and μ). Gray boxes highlight regions with change points identified at consecutive bins. False positive rate: $\alpha = 0.05$.

false positive rate at each time bin. We observe a nonuniform distribution: the false positive rates are lowest near the midpoint but higher toward the start and the end of the trace. This behavior has also been noted in the single-channel version of the algorithm^{12,32} and was understood as arising from the large

uncertainty associated with short segments near the ends. On the other hand, if we calculate the false positive rate of the whole trace, we obtain values of 0.052, 0.050, and 0.048, for one, two, and four channels, respectively. In other words, the desired false positive rate ($\alpha = 0.05$) is indeed approximately achieved

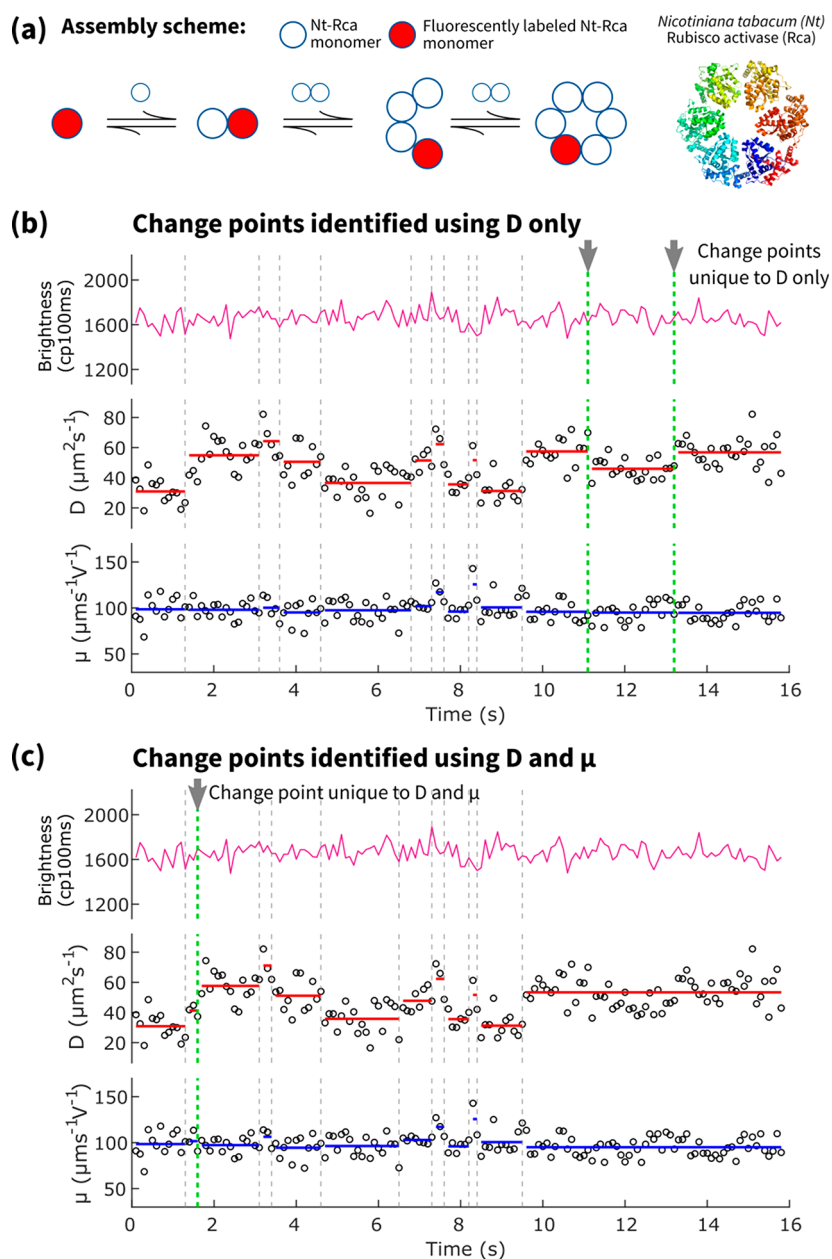


Figure 4. MULLR identifies change points in ABEL trap measurements of Rubisco activase assembly dynamics. (a) Experimental scheme showing a putative Rubisco activase assembly pathway and a hexamer crystal structure (PDB: 3ZW6). (b, c) Example single-molecule trace with change points (gray vertical lines) identified using D only (b) or jointly using D and μ (c). Green vertical dashed lines highlight the differences between the two analyses. Horizontal red and blue lines indicate the mean D and μ values for each level between consecutive change points. False positive rate: $\alpha = 0.05$.

regardless of the number of channels. These results demonstrate that MULLR identifies change points across multiple channels with a well-controlled false positive rate.

MULLR Detects Dynamic Changes in ABEL Trap Measurements. Having validated MULLR on simulated data, we next apply the algorithm to experimental ABEL trap measurements. We first apply MULLR to recent ABEL-FRET data visualizing DNA damage and repair reactions in real time²⁰ (Figure 3a, Figure S4). In this experiment, the smFRET signal, which reports the conformational state of the DNA substrate, is recorded together with single-molecule diffusivity (D) and electrokinetic mobility (μ), which report on protein binding and unbinding. Two distinct DNA-processing proteins are present with the DNA substrate: a nicking endonuclease, which can bind to the dsDNA and cleave one strand, generating a ssDNA

overhang, and a DNA polymerase, which binds and blunts the overhang, regenerating the full dsDNA. We anticipated we would be able to resolve four states using a combination of D and FRET efficiency (Figure 3a).

For this data set, the MULLR algorithm can be applied to different combinations of the four measurement channels to extract different information. When applied to the D and μ channels only (Figure 3b), MULLR segments the data based on the binding state of the DNA substrate. The identified change points agree with visual inspection of the data²⁰ but can now be interpreted as statistically robust and objective identification of state transitions, with a well-defined false positive probability ($\alpha = 0.05$). Further, based on the D and μ values and the smFRET behavior, each segmented level can be assigned to a particular state in the kinetic scheme (circled numbers in panel b). The full

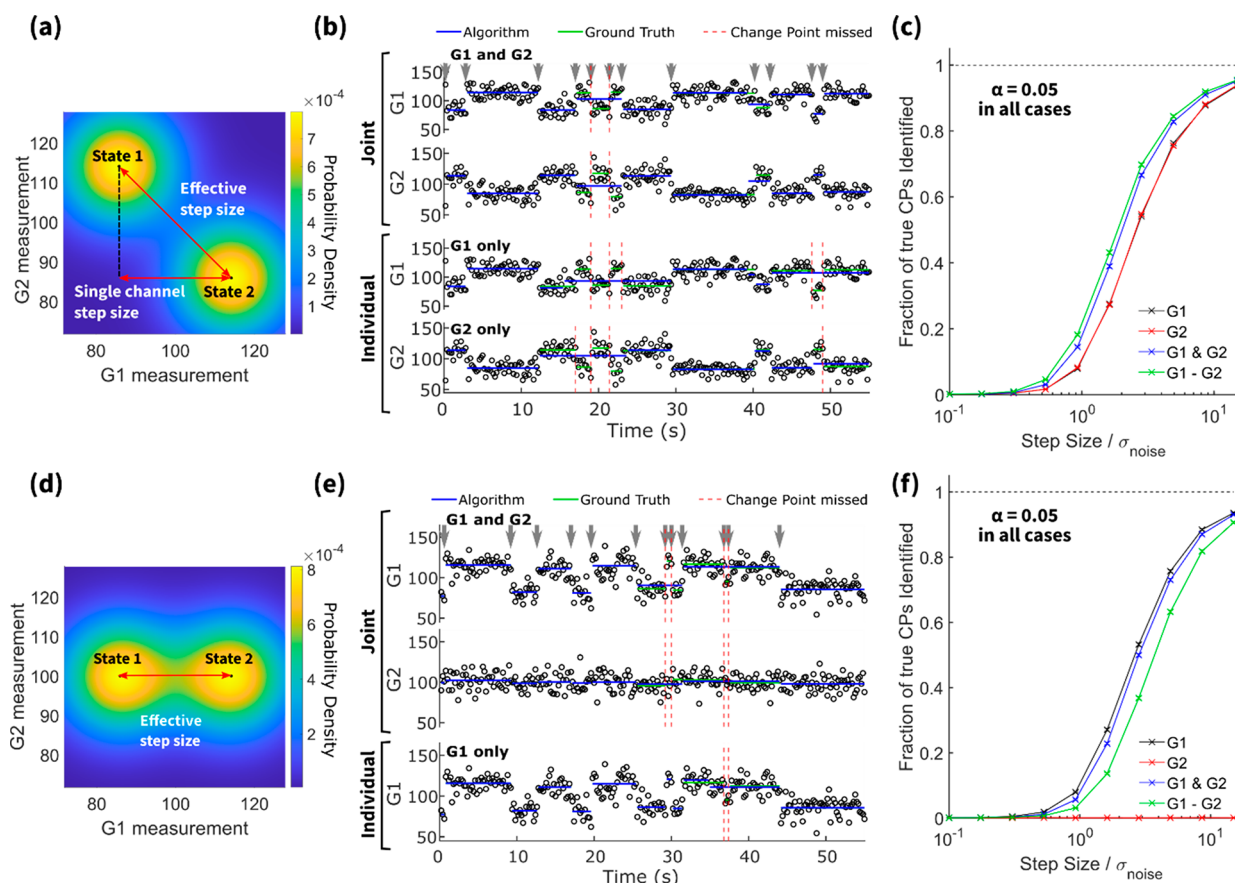


Figure 5. Quantifying the benefit of joint change point analysis: two special cases. (a–c) Synchronous dynamics on two channels with identical step size (case 1). (d–f) Dynamics on only one channel (case 2). For each case, parameter space (G1 and G2) mapping of the two states is shown on the left (panels a and d). The middle panels (b and e) show example simulated traces with the change points identified by joint or individual channel analysis. Gray arrows denote ground truth change point locations. Red dashed lines highlight missed change points. The right panels (c and e) plot the fraction of true change points identified as a function of the single-channel step size, comparing change points identified based on channel 1 only (black), channel 2 only (red), channels 1 and 2 jointly (blue), and channel 1 minus channel 2 (green). Each point is calculated from 1000 simulation repeats. False positive rate: $\alpha = 0.05$.

sequence of molecular events on a single molecule can be inferred.

Applying the MULLR algorithm to all four channels (donor brightness, acceptor brightness, D , and μ) reveals a complete set of dynamic transitions (Figure 3c), including both conformational and binding/unbinding dynamics. Interestingly, for this particular molecule, MULLR returns three regions of change point clusters (i.e., two or more change points at adjacent bins, gray shaded regions, Figure 3c). We believe that those regions correspond to gradual (none stepwise) changes in the binned data, as a result of coarse binning (~ 200 ms) to accommodate the MULLR analysis. We are hopeful that MULLR provides a statistically robust means to analyze future ABEL-FRET data.

We next apply MULLR to analyze the oligomerization dynamics of tobacco Rubisco activase (Rca).³³ In this experiment, a single Rca oligomer undergoes dynamic assembly and disassembly in solution (Figure 4a) and this process is monitored by measuring the motion parameters (diffusivity D and electrokinetic mobility μ) of a single complex in real time with the ABEL trap^{31,33} (Figure 4b). Here, the size-sensitive parameter D undergoes noticeable stepping transitions while the charge-sensitive parameter μ showed much less stepwise variations. We compared change points identified using only D (Figure 4b) with those identified jointly on both D and μ using MULLR (Figure 4c). We find that, although consensus is

reached at many locations, the joint D – μ analysis appears to return fewer change points (notably around 11 and 13 s in Figure 4b and c).

Quantifying the Benefits of Joint Change Point Analysis. The development of MULLR enables us to analyze multichannel data in different ways: we can choose to jointly identify change points across all channels (or a subset of the channels) or run the change point analysis on individual channels.²⁸ Which way is better? As we've seen from the two different types of experiments in the previous section, the optimal choice likely depends on the problem of interest. Next, we attempt to quantify the benefit (or the lack thereof) of joint change point analysis in multichannel data. To this end, we compare the power of detection (under a constant false positive rate) of MULLR to that of single-channel analysis. In this work, we consider two measurement channels with Gaussian noise. We start with two special cases before discussing the more general scenario.

Case 1. Synchronous Dynamics on Two Channels with Identical Step Size. In this case, transitions happen between two states that are positioned diagonally in the parameter space (Figure 5a). We generated 1000 simulated traces and identified change points by analyzing both channels jointly as well as each of the two channels individually (Figure 5b). We plot the fraction of change points identified as a function of the step size

(Figure 5c). We find that the joint analysis captures more change points, with a fractional increase of up to 30% over the single-channel analysis. This can be rationalized by the increased effective separation between the two states in a two-dimensional parameter space compared to the separation along one axis alone (Figure 5a). Indeed, when we replot the power of detection versus the (noise-normalized) effective step size, all curves collapse almost fully onto one another (Figure S5). Interestingly, in this case of perfectly anti-correlated changes, the highest power of detection can be obtained by analyzing the difference of the two channels ($G_1 - G_2$).

Case 2. Dynamics on Only One Channel. Motivated by the Rca experiments (Figure 4), we next consider the case that only G_1 undergoes transitions while G_2 remains constant (a horizontal arrangement in the parameter space, Figure 5d). In this case, joint analysis incurs a small penalty in the power of detection (Figure 5e and f). Here, the effective step size for the joint analysis is the same as that for the single-channel analysis (Figure 5d), but there is a small increase in effective noise due to incorporation of the second channel. Note that, in this case, while the joint approach is slightly inferior to the single-channel approach (with a fractional decrease in power of detection of up to $\sim 10\%$), it is substantially more powerful than the difference channel approach, which performs poorly because it is vulnerable to the noise on the second channel.

Finally, we consider the more general case in which the dynamics are synchronous but the step sizes can take on any pair of values. We simulate two-channel Gaussian time traces with a single synchronous step at the middle and consider all possible combinations of step sizes on the two channels. For each step size combination, we calculate the detection power of the joint analysis using MULLR as well as the detection power when analyzing the single channel with the larger step size. We then compute the difference in detection power (ΔPoD) to quantify the benefit of joint analysis. The result is a color map as a function of the two uncertainty-normalized step sizes (i.e., the step size to uncertainty ratios, $S_{\text{step}}/\sigma_{\text{step}}$, Figure 6). We observe four diagonal regions (red) where the joint analysis gives higher power, and four regions (blue) close to the axes (dashed lines) where the joint analysis resulted in slightly lower power. Interestingly, the highest gain in detection power ($\Delta\text{PoD} \sim 0.3$) is achieved for equal step sizes at a $S_{\text{step}}/\sigma_{\text{step}}$ of ~ 3 . On the other hand, the modest loss in detection power ($\Delta\text{PoD} \sim -0.1$) happens when the steps of the two channels are unequal and of low $S_{\text{step}}/\sigma_{\text{step}}$. When the step size is much bigger than the noise (i.e., $S_{\text{step}}/\sigma_{\text{step}} > 5$), there is no benefit of the joint analysis because the single-channel analysis already has a high enough power of detection.

These results serve as guidance for choosing an optimal analysis strategy for experimental data. To illustrate this, we map the experimentally observed transitions (Figures 3 and 4) in D and μ channels onto Figure 6. For the Nt-Rca assembly experiment (blue crosses), a single-channel analysis on D (the G_1 axis) is expected to give a higher power of detection. For the DNA damage and repair example (black crosses), the joint analysis and the single-channel analysis have almost equivalent power of detection, because of the high $\frac{S_{\text{step}}}{\sigma_{\text{step}}}$ observed on the D and μ channels.

DISCUSSION

In this work, we present an algorithm for joint identification of change points in multichannel single-molecule measurements

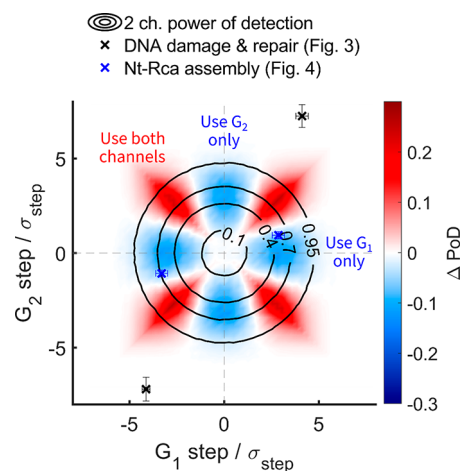


Figure 6. Quantifying the benefit of joint analysis: general case. The color map shows the difference in power of detection between joint and individual channel analysis, for arbitrary step sizes (normalized by the uncertainty in the step size) in two-dimensional parameter space. Black contours are curves of constant power for the joint two-channel analysis. Black crosses map the averaged experimental D (G_1) and μ (G_2) step sizes (normalized) shown in Figure 3 onto the same parameter space; error bars show the standard error of the mean. Blue crosses map the averaged experimental D (G_1) and μ (G_2) step sizes (normalized) shown in Figure 4; error bars show the standard error of the mean. For all analysis, the false positive rate is set at $\alpha = 0.05$.

(MULLR). We build on an existing log-likelihood ratio test and binary segmentation framework. This framework has previously been used extensively to identify change points in single-channel measurements with either Poisson or Gaussian measurement statistics. In our MULLR algorithm, we consider cases with two or more measurement channels, including combinations of channels with different measurement statistics. We characterize the algorithm on simulated data, including its power of detection and false positive rate, and demonstrate applicability on experimental data using the recently developed ABEL-FRET platform. Further, we quantify the potential benefit of joint change point analysis compared to single-channel analysis. We find that, under certain step size combinations and signal-to-noise ratios, which are often encountered in experimental data, joint analysis using MULLR can increase the power of detection (by up to 0.3). On the other hand, single-channel analysis might still be preferred in other cases, notably if only one channel displays dynamic transitions.

Our algorithm inherits the key advantages of the binary segmentation and log-likelihood ratio test framework, which is the minimal number of assumptions we have to make about the model (as discussed in previous work¹²). Another popular class of approaches capable of multichannel analysis is based on the hidden Markov model (HMM) which often assumes a specific model of states and kinetics, or at least some class of models to optimize over.^{7,8,34} This requirement for model specification may become especially problematic as the number of measurement channels increases, because the increased dimensionality of the state space potentially gives a very large space of possibilities to optimize over. Furthermore, the HMM assumes a discrete set of states and exponential dwell times,³⁵ conditions which are not always met in real single-molecule experiments (as previously observed in Li and Yang¹⁴). Accommodating different kinetics requires additional layers of complexity, which will further complicate application of the model. In

contrast, MULLR provides a flexible and straightforward approach to analyzing multichannel data which only requires knowledge of the noise properties of each measurement channel. Recently, deep learning approaches have shown promise in analyzing high dimensional and complex data sets and have been applied to analysis of single-molecule data sets.¹⁷ Future work is required to compare the performance of MULLR to these emergent approaches. However, it is worth noting that MULLR does not require any training sets to operate.

The output change points and levels of the MULLR algorithm developed in this work can be further analyzed using clustering and model selection approaches. These approaches have previously been applied to the one-channel case^{12,14} and would generalize straightforwardly to the multiple-channel case. It would also be possible to expand MULLR to detect changes in parameters other than the mean. For example, changes in the variance of a Gaussian channel could give useful information in studies using tethered particle motion.^{36,37}

We note that our implementation of MULLR also has limitations. The current algorithm requires all measurement channels to have the same time steps and the same number of bins. This may require rebinning the data to the lowest time resolution among the channels. Extensions to analyze multiple channels with different time resolutions (or even photon-by-photon data for brightness channels) will provide a more optimal way to extract fast kinetic information from the data. Future work is also needed to develop and optimize a data analysis pipeline that combines the benefits of single-channel and multichannel analysis.

We envision this work to be a starting point toward more optimal analysis of multichannel single-molecule data that will be useful for a range of hybrid approaches, including multiparameter fluorescence detection^{28,38–40} and the combination of fluorescence with other modalities such as optical tweezers^{19,41–43} and magnetic tweezers.⁴⁴

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.1c08869>.

Supplementary methods; additional figures showing the performance of the MULLR algorithm on simulated and experimental data; and detailed discussion of the algorithm formalism (PDF)

Source code of the MULLR algorithm with tutorial (ZIP)

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Notes

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